

**EBN 111
ELECTRICITY AND
ELECTRONICS**

Chapter 1

Basic Concepts

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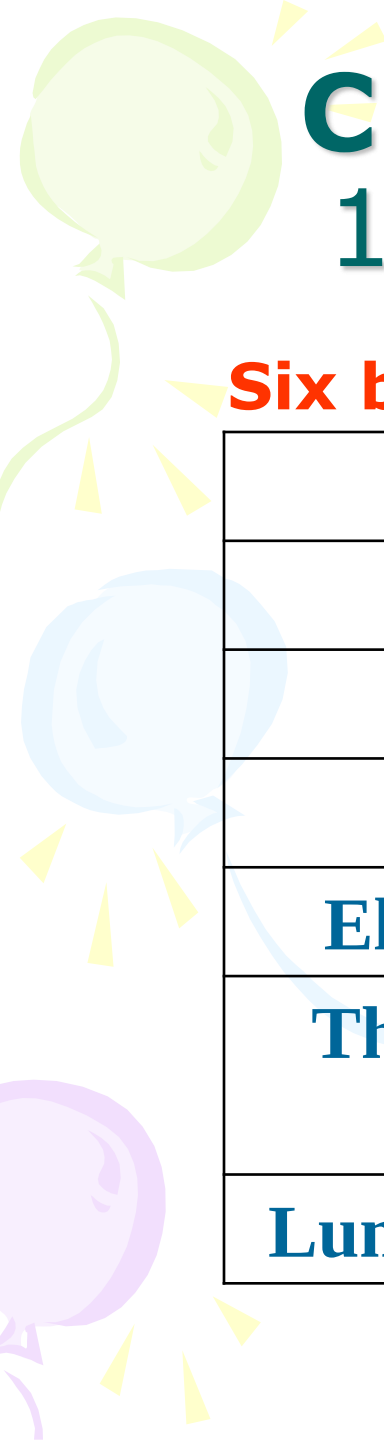
Basic Concepts - Chapter 1

L#1

- 1.1 Systems of Units.
- 1.2 Electric Charge.
- 1.3 Current.
- 1.4 Voltage.

L#2

- 1.5 Power and Energy.
- 1.6 Circuit Elements.



Chapter 1: Lecture #1

1.1 System of Units (1)

Six basic units: SI Units

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd



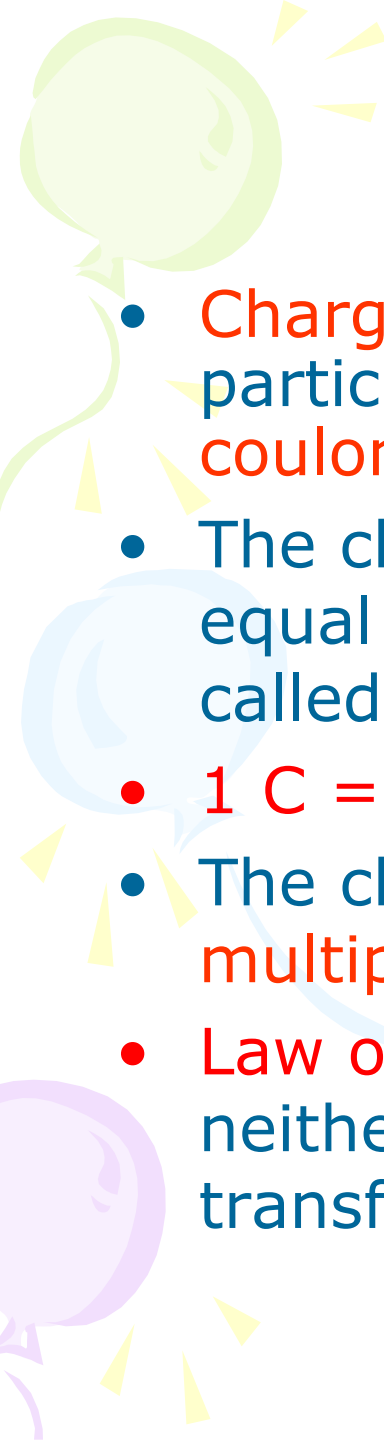
1.1 System of Units (2)

The derived units commonly used in electric circuit theory

**Decimal multiples and
submultiples of SI units**

Quantity	Unit	Symbol
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
conductance	siemens	S
inductance	henry	H
capacitance	farad	F
frequency	hertz	Hz
force	newton	N
energy, work	joule	J
power	watt	W
magnetic flux	weber	Wb
magnetic flux density	tesla	T

Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

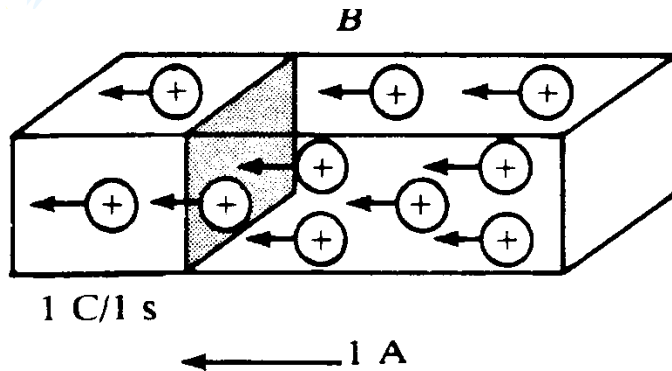


1.2 Electric Charge

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge **e** on one electron is negative and equal in magnitude to **1.602×10^{-19} C** which is called an electronic charge.
- **$1 \text{ C} = 1/e = 6.24 \times 10^{18}$ electrons**
- The charges that occur in nature are **integral multiples** of the electronic charge **e**.
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.

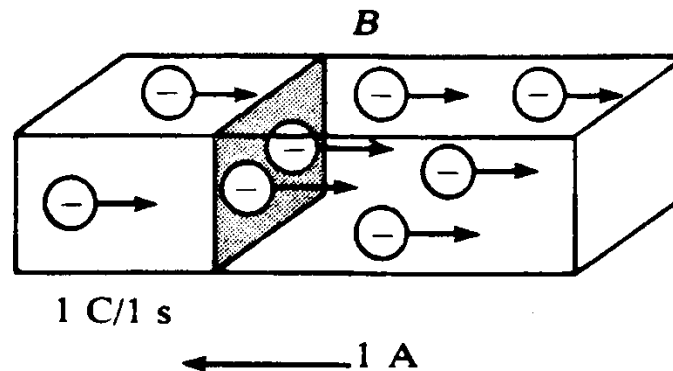
1.3 Current (1)

- The direction of current flow
 - Motion of charge creates electric current.
 - Benjamin Franklin (1706 – 1790): Convention: Current flows in the direction of **positive charge**.
 - Years later it was found that it is actually Free Electrons moving in the opposite direction.



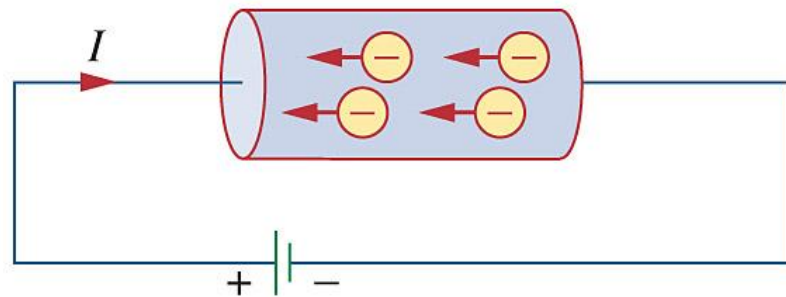
(a)

Positive ions



(b)

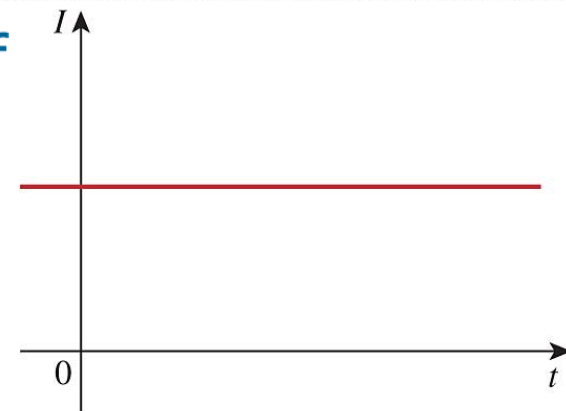
Negative ions



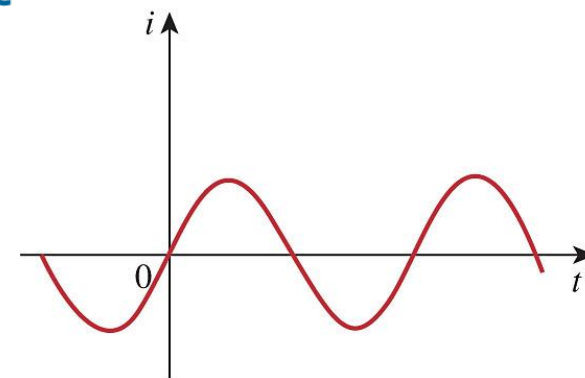
1.3 Current (2)

- Electric current $i = dq/dt$. The unit of ampere can be derived as $1 \text{ A} = 1 \text{ C/s}$.
- Amount of charge transferred between t_0 and t : $Q \triangleq \int_{t_0}^t i \, dt$
- A **direct current (dc)** is a current that remains constant with time.
- An **alternating current (ac)** is a current that varies sinusoidally with time. (reverse direction)

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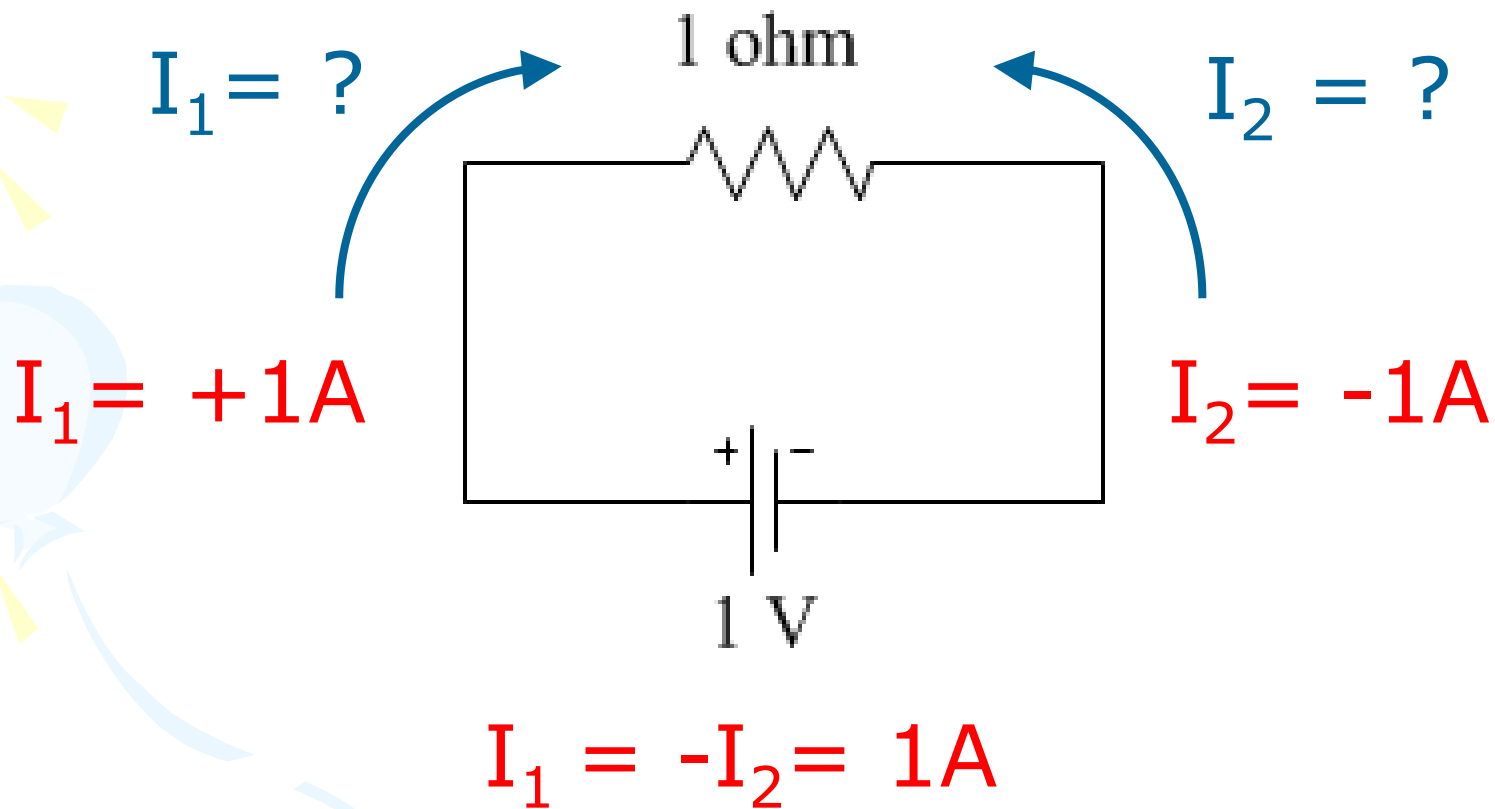


(a)



(b)

1.3 Current (3)



PP1.1

Calculate the amount of charge represented by four million protons.

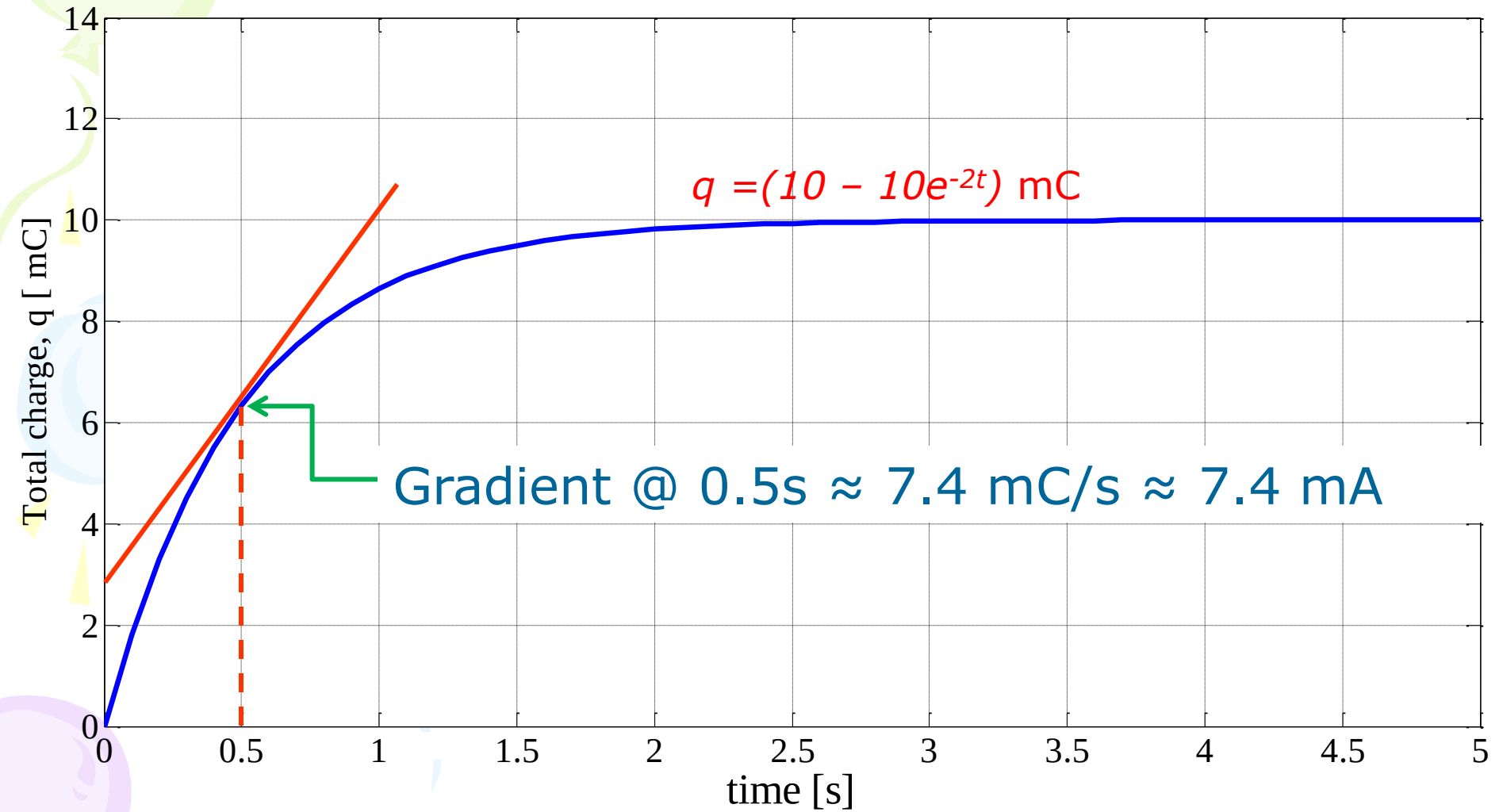
$$\begin{aligned} Q &= 4 \times 10^6 \times -e \\ &= 4 \times 10^6 \times -(-1.602 \times 10^{-19} \text{C}) \\ &= +6.408 \times 10^{-13} \text{C} \\ &= 0.6408 \times 10^{-12} \text{C} \\ &= 0.6408 \text{ pC} \\ &= 640.8 \text{ fC} \end{aligned}$$

PP1-2

The total charge entering a terminal is given by $q = (10 - 10e^{-2t})$ mC. Find the **current** at $t = 0.5s$.

$$i = \frac{dq}{dt}$$

Differentiating means finding the slope or gradient



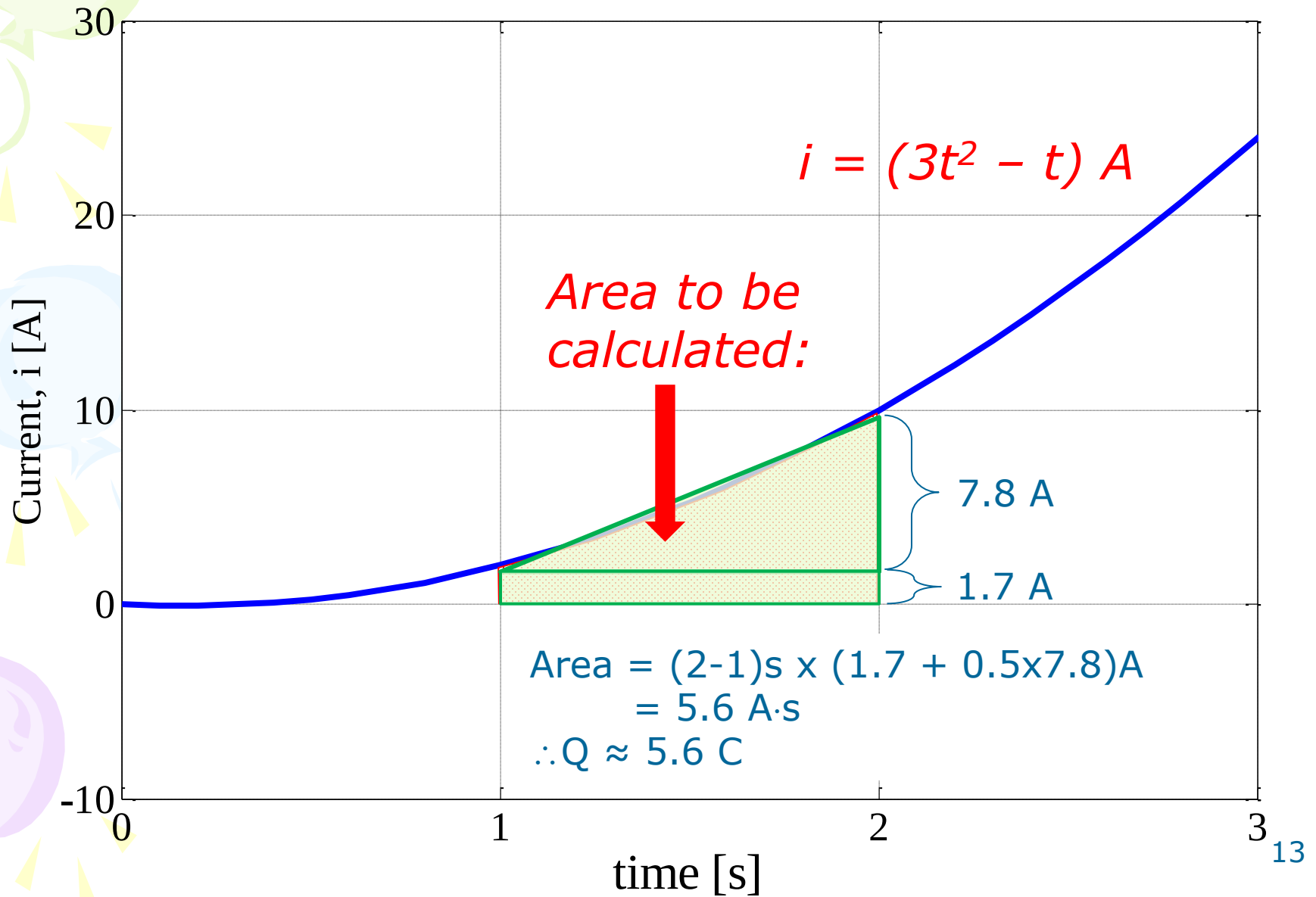
Example 1-3

Determine the total **charge** entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is:

$$i = (3t^2 - t) \text{ A}$$

$$Q = \int_{t_0}^{t_1} i \, dt$$

Integrating means finding the **area** underneath



1.4 Voltage

- Voltage (or potential difference) is the **energy (work)** required to move a **unit charge** through an element, measured in volts (V).
- Work is done by the electromotive force (***emf***) aka ***voltage*** or ***potential difference***.
- Mathematically, (volt)
 - ***w*** is energy in joules (J) and ***q*** is charge in coulomb (C).
 - voltage is change in energy per charge

$$v_{ab} = dw / dq$$

$$V_{ab} = V_a - V_b$$

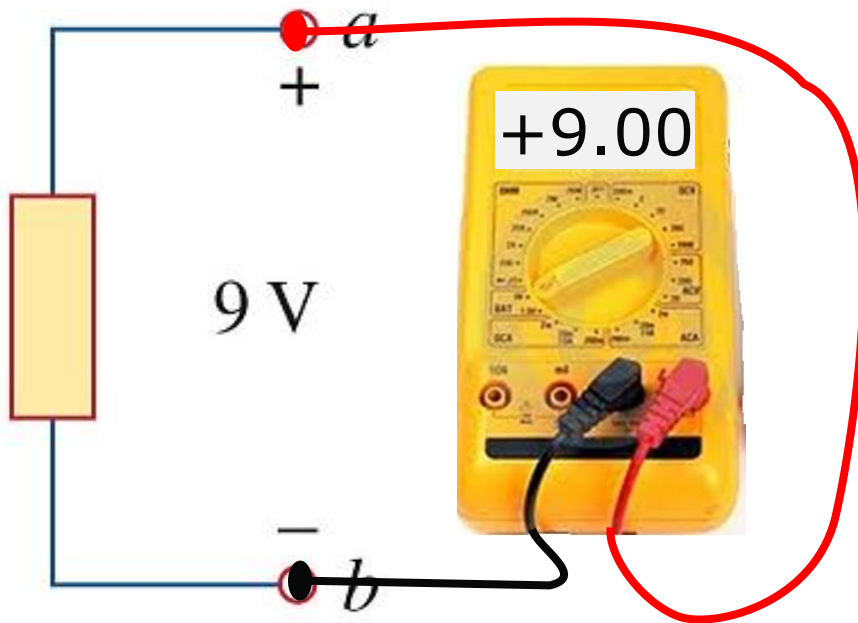
1.4 Voltage

Electric voltage, v_{ab} , is always across the circuit element or between two points in a circuit.

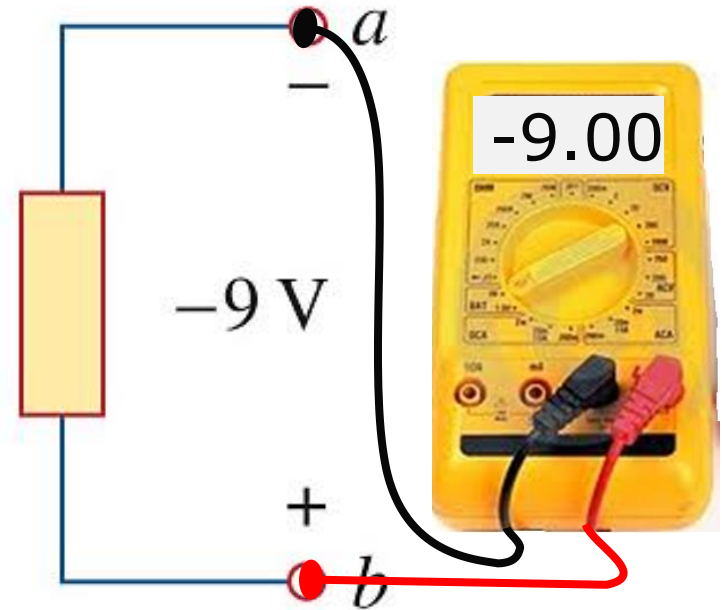
$v_{ab} > 0$ means the potential of **a** is higher than potential of **b**.

$v_{ab} < 0$ means the potential of **a** is lower than potential of **b**.

$$v_{ab} = -v_{ba}$$



(a) $v_{ab} = +9V$



(b) $v_{ba} = -9V$